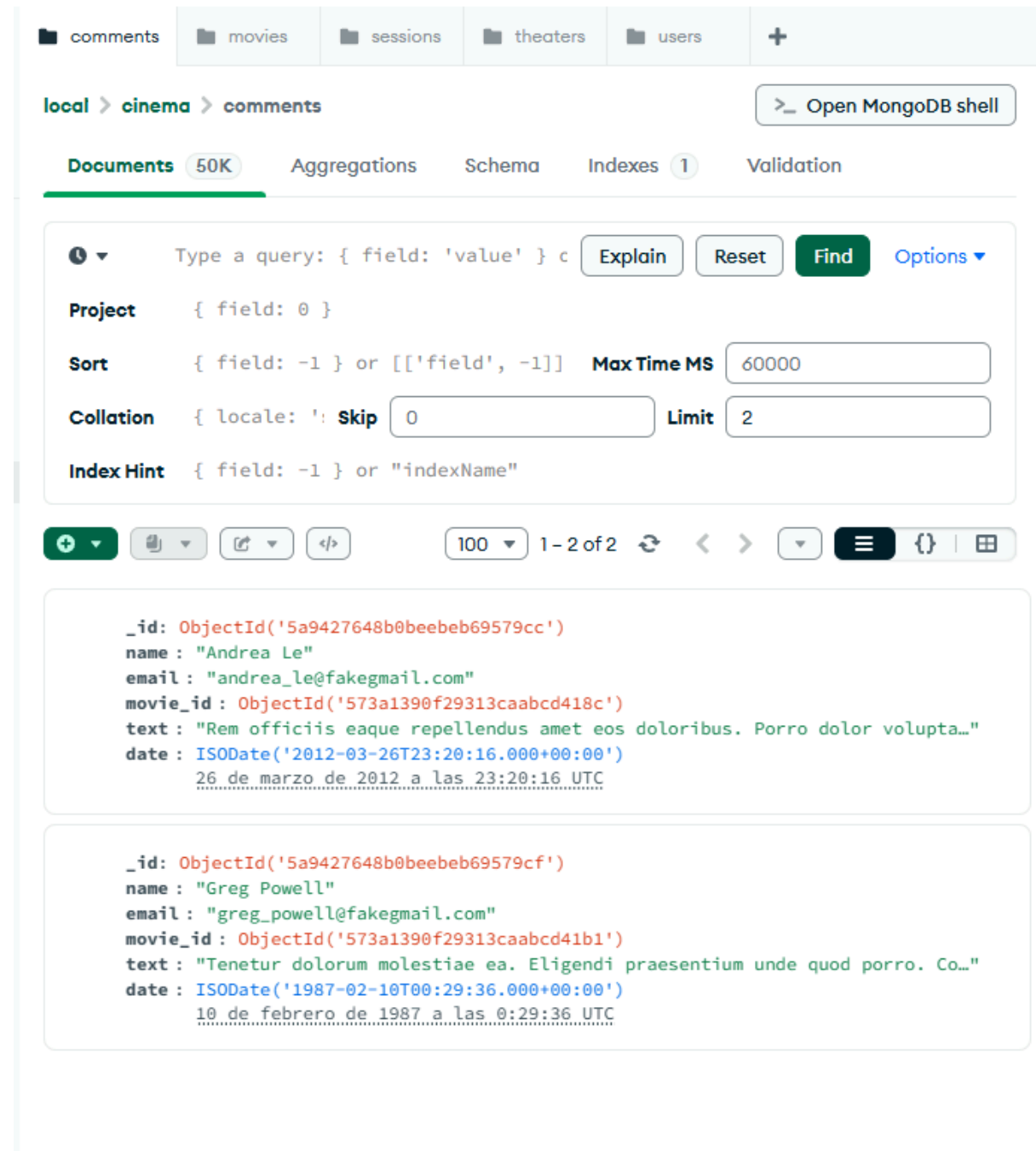


Nivell 1 - Exercici 1

Mostra els 2 primers comentaris que hi ha en la base de dades.

Utilitzem la GUI i posem el valor 2 a la casella de LIMIT.



The screenshot shows the MongoDB GUI interface. At the top, there are tabs for 'comments', 'movies', 'sessions', 'theaters', and 'users'. Below these, the breadcrumb 'local > cinema > comments' is visible. The 'Documents' tab is selected, showing 50K documents. The 'Find' button is highlighted. The query field is empty. The 'Limit' field is set to 2. Two document entries are displayed below the query field.

local > cinema > comments

Documents 50K Aggregations Schema Indexes 1 Validation

Type a query: { field: 'value' } c Explain Reset Find Options

Project { field: 0 }

Sort { field: -1 } or [['field', -1]] Max Time MS 60000

Collation { locale: 'skip' 0 Limit 2

Index Hint { field: -1 } or "indexName"

100 1 - 2 of 2

```
_id: ObjectId('5a9427648b0beebeb69579cc')
name: "Andrea Le"
email: "andrea_le@fakegmail.com"
movie_id: ObjectId('573a1390f29313caab418c')
text: "Rem officiis eaque repellendus amet eos doloribus. Porro dolor volupta..."
date: ISODate('2012-03-26T23:20:16.000+00:00')
      26 de marzo de 2012 a las 23:20:16 UTC
```

```
_id: ObjectId('5a9427648b0beebeb69579cf')
name: "Greg Powell"
email: "greg_powell@fakegmail.com"
movie_id: ObjectId('573a1390f29313caab41b1')
text: "Tenetur dolorum molestiae ea. Eligendi praesentium unde quod porro. Co..."
date: ISODate('1987-02-10T00:29:36.000+00:00')
      10 de febrero de 1987 a las 0:29:36 UTC
```

Quants usuaris tenim registrats?

Sense posar cap filter, ens mostra 185 resultats per usuaris.

The screenshot shows the MongoDB Compass interface. At the top, there are tabs for 'comments', 'movies', 'sessions', 'theaters', and 'users'. The 'users' tab is selected. Below the tabs, the breadcrumb 'local > cinema > users' is visible. To the right, there is a button 'Open MongoDB shell'. Below this, there are tabs for 'Documents', 'Aggregations', 'Schema', 'Indexes', and 'Validation'. The 'Documents' tab is selected, and it shows '185' documents. Below the tabs, there is a search bar with the text 'Type a query: { field: 'value' } c'. To the right of the search bar are buttons 'Explain', 'Reset', 'Find', and 'Options'. Below the search bar, there is a toolbar with icons for adding, deleting, and editing documents, a dropdown for the number of documents (100), and a status bar showing '1 - 100 of 185'.

Quants cinemes hi ha en l'estat de Califòrnia?

Utilitzant el collection de 'theaters', filtrem per location.adress.state: 'CA' i obtenim que hi han 169 cinemes a California.

The screenshot shows the MongoDB Compass interface. At the top, there are tabs for 'comments', 'movies', 'sessions', 'theaters', and 'users'. The 'theaters' tab is selected. Below the tabs, the breadcrumb 'local > cinema > theaters' is visible. To the right, there is a button 'Open MongoDB shell'. Below this, there are tabs for 'Documents', 'Aggregations', 'Schema', 'Indexes', and 'Validation'. The 'Documents' tab is selected, and it shows '1.6K' documents. Below the tabs, there is a search bar with the text '{ "location.address.state": "CA"'. To the right of the search bar are buttons 'Explain', 'Reset', 'Find', and 'Options'. Below the search bar, there is a toolbar with icons for adding, deleting, and editing documents, a dropdown for the number of documents (100), and a status bar showing '1 - 100 of 169'. Below the toolbar, there are three document snippets shown as examples of the filtered results:

```
{
  'location.address.state': 'CA'
}
```

```
{
  _id: ObjectId('59a47286cfa9a3a73e51e72e')
  theaterId: 1008
  location: Object (2)
}
```

```
{
  _id: ObjectId('59a47286cfa9a3a73e51e734')
  theaterId: 1009
  location: Object (2)
}
```

```
{
  _id: ObjectId('59a47286cfa9a3a73e51e737')
  theaterId: 1018
  location: Object (2)
}
```

Quin va ser el primer usuari/ària en registrar-se?

Com que no tenim data d'enregistrament, hem d'assumir que el primer usuari de la nostra collection és l'usuari que es va enregistrar primer, és a dir, Ned Stark va ser el primer usuari.

comments

movies

sessions

theaters

users

+

local > cinema > users

> Open MongoDB shell

Documents 185 Aggregations Schema Indexes 1 Validation

⌚

Type a query: { field: 'value' } c

Explain

Reset

Find

Options

Project

{ field: 0 }

Sort

{ field: -1 } or [['field', -1]]

Max Time MS

60000

Collation

{ locale: '': Skip 0

Limit

1

Index Hint

{ field: -1 } or "indexName"

+

📄

🔗

</>

100

1 - 1 of 1

↺

↻

↷

☰

{ }

🔍

_id: ObjectId('59b99db4cfa9a34dcd7885b6')

name : "Ned Stark"

email : "sean_bean@gameofthron.es"

password : "\$2b\$12\$UREFwsRUoyF0CRqGNK0Lz00HM/jLhgUCNNIJ9RJAqMUQ74cr1J1Vu"

Quantes pel·lícules de comèdia hi ha en la nostra base de dades?

Hi ha 7024 pel·lícules de comèdia.

Filter

```
{
  genres: 'Comedy'
}
```

comments

movies

sessions

theaters

users

+

local > cinema > movies

> Open MongoDB shell

Documents 24K

Aggregations

Schema

Indexes 1

Validation



{ genres: "Comedy" }



Explain

Reset

Find

Options ▶



1... ▼

1 - 100 of
7024

```
_id: ObjectId('573a1390f29313caabcd4803')
plot: "Cartoon figures announce, via comic strip balloons, that they will mov..."
▶ genres: Array (3)
  runtime: 7
▶ cast: Array (1)
  num_mflix_comments: 1
  poster: "https://m.media-amazon.com/images/M/MV5BYzg2NjNhNTctMjUxMi00ZWU4LWI3Zj..."
  title: "Winsor McCay, the Famous Cartoonist of the N.Y. Herald and His Moving ..."
  fullplot: "Cartoonist Winsor McCay agrees to create a large set of drawings that ..."
▶ languages: Array (1)
  released: ISODate('1911-04-08T00:00:00.000+00:00')
           8 de abril de 1911 a las 0:00:00 UTC
▶ directors: Array (2)
▶ writers: Array (2)
▶ awards: Object (3)
  lastupdated: "2015-08-29 01:09:03.030000000"
  year: 1911
▶ imdb: Object (3)
▶ countries: Array (1)
  type: "movie"
▶ tomatoes: Object (2)
```

```
_id: ObjectId('573a1390f29313caabcd50e5')
plot: "The cartoonist, Winsor McCay, brings the Dinosaurus back to life in th..."
▶ genres: Array (3)
  runtime: 12
▶ cast: Array (3)
```

Nivell 1 - Exercici 2

Per mostrar els documents de les pel·lícules produïdes en 1932, però que el gènere sigui drama o estiguin en francès, hem d'utilitzar el filtre `{condició1,$or[condició2, condició3]}`:

Filter

```
{
  year: 1932,
  $or: [
    {
      genres: 'Drama'
    },
    {
      languages: 'French'
    }
  ]
}
```

comments movies sessions theaters users +

local > cinema > movies [Open MongoDB shell](#)

Documents 24K Aggregations Schema Indexes 1 Validation

[Filter](#) {year:1932,\$or:[{genres:'Drama'}]} [Explain](#) [Reset](#) [Find](#) [Options](#)

100 1 - 18 of 18

```
_id: ObjectId('573a1391f29313caabcd9458')
plot: "A young artist draws a face at a canvas on his easel. Suddenly the mou..."
runtime: 55
rated: "UNRATED"
cast: Array (4)
num_mflix_comments: 1
poster: "https://m.media-amazon.com/images/M/MV5BYWY3ODE5ZWEtYjlmYi00NjA4LTk4ZW..."
title: "The Blood of a Poet"
lastupdated: "2015-09-16 13:13:05.537000000"
languages: Array (1)
  0: "French"
released: ISODate('2010-05-20T00:00:00.000+00:00')
  20 de mayo de 2010 a las 0:00:00 UTC
directors: Array (1)
writers: Array (1)
awards: Object (3)
  year: 1932
imdb: Object (3)
countries: Array (1)
  type: "movie"
tomatoes: Object (7)
```

```
_id: ObjectId('573a1392f29313caabcd99a3')
plot: "Junta is hated by the people in the village where she lives, especiall..."
genres: Array (3)
  0: "Drama"
  1: "Fantasy"
  2: "Mystery"
runtime: 85
```

Nivell 1 - Exercici 3

Mostrem tots els documents de pel·lícules amb país USA que tinguin entre 5 i 9 premis que van ser produïdes entre 2012 i 2014.

Filter

```
{
  countries: 'USA',
  year: {
    $gte: 2012,
    $lte: 2014
  },
  'awards.wins': {
    $gte: 5,
    $lte: 9
  }
}
```

local > cinema > movies [Open MongoDB shell](#)

Documents 24K Aggregations Schema Indexes 1 Validation

[🕒](#) [🔍](#) {countries:'USA',year:{\$gte:2012,\$lte:2014,'awards.wins':{\$gte:5,\$lte:9}}} [Explain](#) [Reset](#) [Find](#) [Options ▶](#)

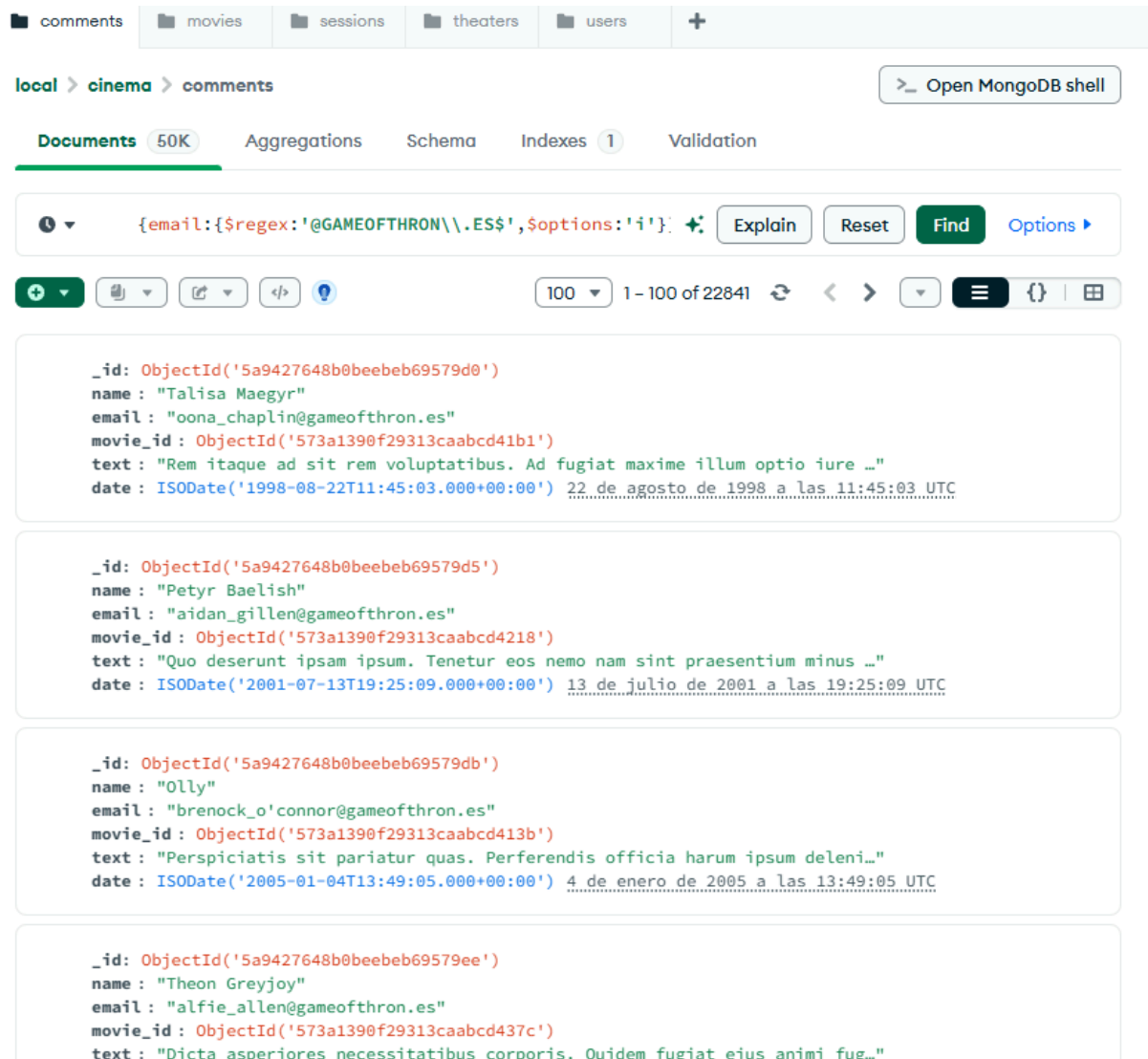
[+](#) [📄](#) [🔗](#) [🔧](#) [💡](#) 100 1 - 100 of 166 [↺](#) [↻](#) [↷](#) [☰](#) [{ }](#) [🏠](#)

```
_id: ObjectId('573a13acf29313caabd29366')
fullplot: "The manager of the negative assets sector of Life magazine, Walter Mit..."
imdb: Object (3)
year: 2013
plot: "When his job along with that of his co-worker are threatened, Walter t..."
genres: Array (3)
rated: "PG"
metacritic: 54
title: "The Secret Life of Walter Mitty"
lastupdated: "2015-08-31 00:10:51.747000000"
languages: Array (3)
writers: Array (3)
type: "movie"
tomatoes: Object (10)
poster: "https://m.media-amazon.com/images/M/MV5BODYwNDYxNDk1Nl5BMl5BanBnXkFtZT..."
num_mflix_comments: 1
released: ISODate('2013-12-25T00:00:00.000+00:00')
          25 de diciembre de 2013 a las 0:00:00 UTC
awards: Object (3)
countries: Array (2)
cast: Array (4)
directors: Array (1)
runtime: 114
```

```
_id: ObjectId('573a13b5f29313caabd45772')
plot: "After their cave is destroyed, a caveman family must trek through an u..."
genres: Array (3)
runtime: 98
```

Nivell 2 - Exercici 1

Aquesta sentència regex buscarà tots els email que acabin en '@GAMEOFTHRON.ES' tant en minúscula com en majúscula degut a la opció 'i'.



The screenshot shows the MongoDB Compass interface. At the top, there are tabs for 'comments', 'movies', 'sessions', 'theaters', and 'users'. Below these, the breadcrumb 'local > cinema > comments' is visible. A button 'Open MongoDB shell' is on the right. The main section has tabs for 'Documents' (50K), 'Aggregations', 'Schema', 'Indexes' (1), and 'Validation'. A search bar contains the regex query: `{email:{$regex:'@GAMEOFTHRON\\.\.ES$',options:'i'}}`. Below the search bar, there are four document results, each with fields: `_id`, `name`, `email`, `movie_id`, `text`, and `date`. The results are as follows:

Document	name	email	movie_id	text	date
1	Talisa Maegyr	oona_chaplin@gameofthron.es	ObjectId('573a1390f29313caabcd41b1')	Rem itaque ad sit rem voluptatibus. Ad fugiat maxime illum optio iure ...	22 de agosto de 1998 a las 11:45:03 UTC
2	Petyr Baelish	aidan_gillen@gameofthron.es	ObjectId('573a1390f29313caabcd4218')	Quo deserunt ipsam ipsum. Tenetur eos nemo nam sint praesentium minus ...	13 de julio de 2001 a las 19:25:09 UTC
3	Ollly	brenock_o'connor@gameofthron.es	ObjectId('573a1390f29313caabcd413b')	Perspiciatis sit pariatur quas. Perferendis officia harum ipsum delenit...	4 de enero de 2005 a las 13:49:05 UTC
4	Theon Greyjoy	alfie_allen@gameofthron.es	ObjectId('573a1390f29313caabcd437c')	Dicta asperiores necessitatibus corporis. Quidem fugiat eius animi fug...	

Nivell 2 - Exercici 2

Per fer aquest exercici, necessitem utilitzar una agregació de dues etapes:

1 - Utilitzem el \$match per filtrar el resultats a cinemes de l'estat de DC.

▼ Stage 1 \$match 

```
1 {  
2   "location.address.state": "DC"  
3 }
```

2 - Utilitzem el \$group per agrupar els resultat per codi postal.

▼ Stage 2 \$group 

```
1 {  
2   _id: "$location.address.zipcode",  
3   count: {  
4     $sum: 1  
5   }  
6 }
```

Després, executem el codi i podem observar que cada codi postal de l'estat de Washington DC té exactament 1 cinema.

ALL RESULTS

 EXPORT DATA

 EXPORT CODE

```
_id: "20016"  
count : 1
```

```
_id: "20010"  
count : 1
```

```
_id: "20002"  
count : 1
```

Nivell 3 - Exercici 1

Troba totes les pel·lícules dirigides per John Landis amb una puntuació IMDb (Internet Movie Database) d'entre 7,5 i 8.

Filter

```
{
  directors: 'John Landis',
  'imdb.rating': {
    $gte: 7.5,
    $lte: 8
  }
}
```

Project

```
{
  _id: 0,
  title: 1
}
```

comments

movies

sessions

theaters

users

+

local > cinema > movies

Open MongoDB shell

Documents 24K Aggregations Schema Indexes 1 Validation

🕒 {directors:'John Lan ⚙️ Explain Reset Find Options ▾

Project { _id: 0, title:1}

Sort { field: -1 } or [['fi Max Time MS 60000

Collation { locale: Skip 0 Limit 0

Index Hint { field: -1 } or "indexName"

🔗 </> ? 100 1 - 4 of 4 ↺ < > ▾ ☰ {} | 🏠

title: "Animal House"

title: "The Blues Brothers"

title: "An American Werewolf in London"

title: "Trading Places"

Nivell 3 - Exercici 2

Per mostrar en un mapa totes les ubicacions dels cinemes de la nostra base de dades, simplement hem d'anar a la collection de 'theaters', fer click en 'schema' i fer scroll fins arribar a les dades de localització 'geo' i desplegar el menú, el que ens mostrarà un mapa ja que les dades 'geo' son de longitud/latitud i MongoDB Compass ho detecta automàticament

comments

movies

sessions

theaters

users

+

local > cinema > theaters

Open MongoDB shell

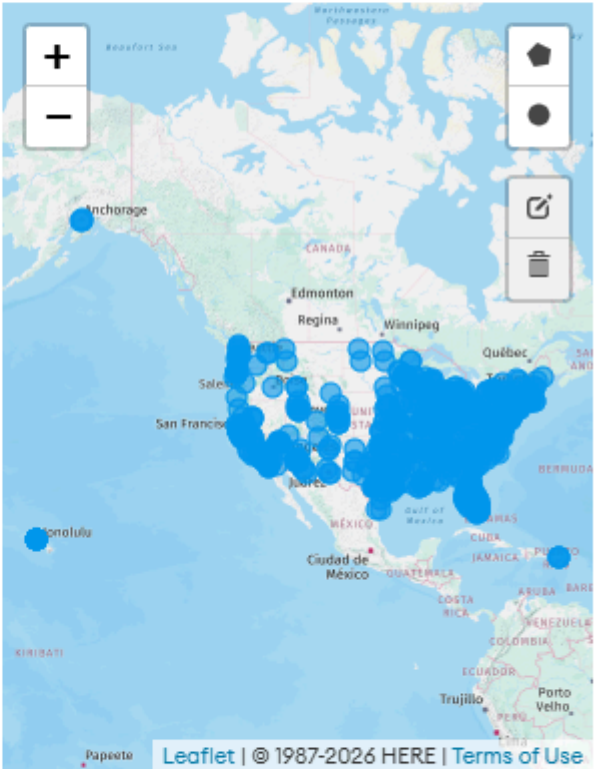
Documents 1.6K Aggregations Schema Indexes 1 Validation

Type a query: { field: 'value' } Reset Analyze Options

EXPORT SCHEMA This report is based on a sample of 1000 documents. [Learn more](#)

geo

coordinates



theaterId

int32

1166

236

8025

2820

8173

594

8046

183

851

1137

1444

1507

